Consequences

From the conditional statement $p \to q$, three more conditional statements are derived. They are listed below.

(i) Converse statement $q \to p$.

(ii) Inverse statement $\neg p \to \neg q$.

(iii) Contrapositive statement $\neg q \to \neg p$.

Example 12.15

Write down the (i) conditional statement (ii) converse statement (iii) inverse statement, and (iv) contrapositive statement for the two statements p and q given below.

p : The number of primes is infinite. q: Ooty is in Kerala.

Solution

Then the four types of conditional statements corresponding to p and q are respectively listed below.

(i) $p \to q$: (conditional statement) "**If** the number of primes is infinite **then** Ooty is in Kerala".

(ii) $q \to p$: (converse statement) "**If** Ooty is in Kerala **then** the number of primes is infinite"

(iii) $\neg p \to \neg q$ (inverse statement) "**If** the number of primes is **not** infinite **then** Ooty is **not** in Kerala".

(iv) $\neg q \to \neg p$ (contrapositive statement) "**If** Ooty is **not** in Kerala **then** the number of primes is **not** infinite".

Bi-conditional Statement

Definition 12.14

The **bi-conditional statement** of any two statements p and q is the statement "p if and only if q" and is denoted by $p \leftrightarrow q$. Its truth value is T, whenever both p and q have the same truth values, otherwise it is false.

Truth table for $p \leftrightarrow q$

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Table 12.8

Exclusive OR (EOR)[$\bar{\vee}$]

Definition 12.15

Let p and q be any two statements. Then p **EOR** q is such a compound statement that its truth value is decided by either p or q but **not both**. It is denoted by $p \,\bar{\vee}\, q$. The truth value of $p \,\bar{\vee}\, q$ is T whenever either p or q is T, otherwise it is F. The truth table of $p \,\bar{\vee}\, q$ is given below.

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